

SnpEff: Variant analysis**Contents**

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Summary

Genome	Nipponbare
Date	2025-09-02 10:02
SnpEff version	SnpEff 5.2 (build 2023-09-29 06:17), by Pablo Cingolani
Command line arguments	SnpEff -stats ML1_snpEff_summary.html Nipponbare ML-1_filtered_2_renamed.vcf.gz
Warnings	5,152,951
Errors	68
Number of lines (input file)	3,778,533
Number of variants (before filter)	3,778,533
Number of not variants (i.e. reference equals alternative)	0
Number of variants processed (i.e. after filter and non-variants)	3,778,533
Number of known variants (i.e. non-empty ID)	0 (0%)
Number of multi-allelic VCF entries (i.e. more than two alleles)	0
Number of effects	10,969,683
Genome total length	385,710,681
Genome effective length	385,710,681
Variant rate	1 variant every 102 bases

Variants rate details

Chromosome	Length	Variants	Variants rate
1	43,929,697	418,680	104
2	36,447,916	359,709	101
3	37,399,924	357,357	104
4	36,078,568	323,568	111
5	30,400,764	255,160	119
6	32,122,276	307,838	104
7	29,936,421	314,257	95
8	28,605,474	297,512	96
9	27,474,823	240,669	114
10	23,931,887	277,600	86
11	31,111,469	349,117	89
12	28,271,460	276,998	102
NC_001751.1	1	0	0
NC_011033.1	1	0	0
Total	385,710,681	3,778,533	102

Number variants by type

Type	Total
SNP	3,411,146
MNP	0
INS	177,275
DEL	190,112
MIXED	0
INV	0
DUP	0
BND	0
INTERVAL	0
Total	3,778,533

Number of effects by impact

Type (alphabetical order)	Count	Percent
HIGH	41,829	0.381%
LOW	253,544	2.311%
MODERATE	375,256	3.421%
MODIFIER	10,298,986	93.886%

Number of effects by functional class

Type (alphabetical order)	Count	Percent
MISSENSE	375,459	60.613%
NONSENSE	17,590	2.84%
SILENT	226,385	36.547%

Missense / Silent ratio: 1.6585

Number of effects by type and region

Type			Region		
Type (alphabetical order)	Count	Percent	Type (alphabetical order)	Count	Percent
3_prime_UTR_variant	97,524	0.886%	DOWNSTREAM	3,351,792	30.555%
5_prime_UTR_premature_start_codon_gain_variant	6,852	0.062%	EXON	636,877	5.806%
5_prime_UTR_variant	60,526	0.55%	INTERGENIC	2,479,231	22.601%
conservative_inframe_deletion	3,435	0.031%	INTRON	730,429	6.659%
conservative_inframe_insertion	3,167	0.029%	SPLICE_SITE_ACCEPTOR	2,060	0.019%
disruptive_inframe_deletion	2,028	0.018%	SPLICE_SITE_DONOR	1,964	0.018%
disruptive_inframe_insertion	1,741	0.016%	SPLICE_SITE_REGION	22,924	0.209%
downstream_gene_variant	3,351,804	30.464%	TRANSCRIPT	498	0.005%
exon_loss_variant	3	0%	UPSTREAM	3,578,964	32.626%
frameshift_variant	9,753	0.089%	UTR_3_PRIME	97,513	0.889%
initiator_codon_variant	71	0.001%	UTR_5_PRIME	67,363	0.614%
intergenic_region	2,479,231	22.534%			
intragenic_variant	267	0.002%			
intron_variant	753,496	6.849%			
missense_variant	365,231	3.32%			
non_coding_transcript_variant	231	0.002%			
splice_acceptor_variant	2,111	0.019%			
splice_donor_variant	2,032	0.018%			
splice_region_variant	28,783	0.262%			
start_lost	1,340	0.012%			
start_retained_variant	19	0%			
stop_gained	17,887	0.163%			
stop_lost	9,418	0.086%			
stop_retained_variant	2,732	0.025%			
synonymous_variant	223,682	2.033%			
upstream_gene_variant	3,578,964	32.529%			



Quality:

Min	30
Max	228
Mean	199.227
Median	225
Standard deviation	53.262
Values	30,31,32,33,34,35,36,37,38,39,40,41,42,43,44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59,60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75,76
Count	6944,6737,6621,6658,6445,6277,6465,7507,6079,6365,6292,6178,6302,6086,6436,5938,5916,5900,5902,5925,6039,5760,5815,5570,5715,5746,



Insertions and deletions length:

Min	0
Max	44
Mean	1.887
Median	1
Standard deviation	3.897
Values	0,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,44
Count	86456,218240,14951,9508,4554,5506,3112,3440,2829,2130,1838,2214,1234,1184,1183,931,830,895,696,624,603,536,479,477,355,323,321,285,1



Base changes (SNPs)

	A	C	G	T
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A	0	122,770	558,845	138,325
C	133,672	0	93,287	657,728
G	657,920	93,683	0	132,830
T	138,977	561,148	121,961	0

Ts/Tv (transitions / transversions)

Note: Only SNPs are used for this statistic.
Note: This Ts/Tv ratio is a 'raw' ratio (ratio of observed events).

Transitions	4,294,940
Transversions	1,721,462
Ts/Tv ratio	2.4949

All variants:

Sample ,ML-1_sorted.bam,Total
Transitions ,4294940,4294940
Transversions ,1721462,1721462
Ts/Tv ,2.495,2.495

Only known variants (i.e. the ones having a non-empty ID field):

No results available (empty input?)

Allele frequency



Min	50
Max	100
Mean	88.736
Median	100
Standard deviation	20.889
Values	50,100
Count	851254,2927279

Allele Count



Min	1
Max	2
Mean	1.775
Median	2
Standard deviation	0.418
Values	1,2
Count	851254,2927279

Hom/Het per sample



Sample_names , ML-1_sorted.bam
Reference , 0
Het , 851254
Hom , 2927279
Missing , 0

Codon changes

How to read this table:
- Rows are reference codons and columns are changed codons. E.g. Row 'AAA' column 'TAA' indicates how many 'AAA' codons have been replaced by 'TAA' codons.
- Red background colors indicate that more changes happened (heat-map).
- Diagonals are indicated using grey background color
- WARNING: This table may include different translation codon tables (e.g. mamalian DNA and mitochondrial DNA).

	-	AAA	AAC	AAG	AAN	AAT	ACA	ACC	ACG	ACT	AGA	AGC	AGG	AGN	AGT	ATA	ATC	ATG	ATT	CAA	CA
-		122	147	313		121	129	229	217	112	191	216	259	1	106	89	143	208	76	170	1
AAA	380	68	433	2,704		448	366	2		1	1,507	3	6		1	260	1	1	2	582	
AAC	348	483	77	525		3,680	5	381	3	1	3	1,891	1		2	1	322	1		1	3
AAG	662	3,218	613	120		843	2	3	408	1	4	4	2,143		2	5	3	690	1	3	
AAN						1															
AAT	306	468	3,071	631		52	3	1	1	327	5	2	1		1,401	1	1		337	1	
ACA	311	466	11	1		1	69	694	2,825	582	412	5	2		3	2,313	1	1	2		
ACC	455		382	2		2	692	137	480	2,670		399	2		5		1,536	4	1	1	
ACG	403	1		420		1	3,316	417	113	454	1	3	348				3	4,142			
ACT	221	1	4			360	630	2,252	418	53	2	2	2		372	5	1	3	1,544	1	
AGA	408	1,868	2	22		2	411	1	3	1	63	440	1,802		317	589	1	3		1	
AGC	440	1	2,152	8		1	1	395	2		482	104	345		2,554	1	409	2	1		
AGG	565	7	3	2,302		2	3		337		2,069	360	96		366		3	593	1		
AGN										1					1						

	-	AAA	AAC	AAG	AAN	AAT	ACA	ACC	ACG	ACT	AGA	AGC	AGG	AGN	AGT	ATA	ATC	ATG	ATT	CAA	CA
AGT	234			9		1,526	4	3	5	387	324	2,130	313		45	3		2	312	1	
ATA	203	277	1			4	1,722			4	410				1	27	602	1,411	461		
ATC	328		305			4	1	1,333		3		321	1			698	47	418	2,804	1	
ATG	538	1		704		12	3	6	4,083	5	10	2	525		3	2,153	486	146	549	2	
ATT	253	2		1		358				1,320	1	1			260	415	2,303	435	41	1	
CAA	348	770				1	3					1								91	3
CAC	362		370				1	1												468	10
CAG	611			667			2		1		1									3,971	4
CAT	278					366	4									1		1		428	2,2
CCA	439		1				421		2			1	1		1				1	608	
CCC	393		1					298		1			1				1			2	3
CCG	747								328				2							1	
CCT	381					1		1	1	378											
CGA	280							1			534		2							3,032	
CGC	583			2				1	3			457									3,7
CGG	567								2				695							2	
CGT	209								1	1					270					1	
CTA	197									1			1			330				410	
CTC	554		1									1					563	1		1	4
CTG	439			1			1		1	4	1						1	740		2	
CTT	331		1					2										1	496	4	
GAA	479	3,059	1	1		1					2				1			1		580	
GAC	807		2,762	8							2	1	1					2			3
GAG	1,039		1	3,444					1		7	1			1		1	2		2	
GAN																					
GAT	439		1	1		2,515					1	1	1				5	5	1		
GCA	485	1				1	2,732	1	3		1	1	1					2			
GCC	1,383		1	1			3	3,603	2	1		11	1				2	1			
GCG	1,567			1			1	1	2,884				2				2	2			
GCT	471		1	1			1	1		2,418	2	3						2			
GGA	594		1			2			1		2,960		7								
GGC	1,747		1	1						1		3,480	6								
GGG	653	1	1	2									2,353					1		1	
GGT	433			1				2		1	1				2,000						
GNN																					
GTA	172														2	1,398					
GTC	470			1											1		3,193	1			
GTG	513		1						1			1			1	1	1	2,840			
GTT	291	1		2		1									2			1	2,579	2	
TAA	120	136														2				881	
TAC	257	1	303	2				1				1			1	1					9
TAG	92			152							1				1					1	
TAT	195					264											1				
TCA	307					1	388				1	1	1				2				
TCC	464		2					430	1		1				2		3		1		
TCG	428							1	338				2				3				
TCT	380					1				386			2				5	1	1		
TGA	219								1		248					1	1	2			
TGC	306		1			1						372						3			
TGG	405					1					1		471		1			1		1	
TGT	208						1		1						335			4			
TTA	171					1									1	240			1		
TTC	389					1											366	1			
TTG	350			1		1				1								541	2		
TTT	281														1				362		

Amino acid changes

How to read this table:

- Rows are reference amino acids and columns are changed amino acids. E.g. Row 'A' column 'E' indicates how many 'A' amino acids have been replaced by 'E' amino acids.
- Red background colors indicate that more changes happened (heat-map).
- Diagonals are indicated using grey background color
- WARNING: This table may include different translation codon tables (e.g. mamalian DNA and mitochondrial DNA).

	*	-	?	A	C	D	E	F	G	H	I	K	L	M	N	P	Q	R	S	T
*	2,793	428	3		375	1	336	2	280	1	4	288	580	2		4	2,168	2,274	629	
-	229		4,240	2,375	211	650	804	256	2,021	312	308	435	881	208	268	1,222	512	1,376	1,165	68
?		63	277			4			1						1				1	
A		3,905	1	26,304	7	1,001	1,338		2,179	3	4	4	16	7	3	1,589	3	27	2,574	11,65
C	458	514		1	4,496		2	754	755	2			24	7	2	6		4,487	1,326	
D	10	1,245	1	901		9,950	2,799	1	3,382	755	6	9	2	7	5,278	2	2	7	5	
E	939	1,517	1	1,283		2,948	6,568	1	3,100		1	6,504	2	3	3	4	1,262	15	4	

[illegible]

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12, Position,0,100000,200000,300000,400000,500000,600000,700000,800000,900000,1000000,1100000,1200000,1300000,1400000,1500000,1600000,1700000,1800000,1900000,2000000,2100000,2200000,2300000,2400000,2500000,2600000,2700000,2800000,2900000,3000000,3100000,3200000,3300000,3400000,3500000,3600000,3700000,3800000,3900000,4000000,4100000,4200000,4300000,4400000,4500000,4600000,4700000,4800000,4900000,5000000,5100000,5200000,5300000,5400000,5500000,5600000,5700000,5800000,5900000,6000000,6100000,6200000,6300000,6400000,6500000,6600000,6700000,6800000,6900000,7000000,7100000,7200000,7300000,7400000,7500000,7600000,7700000,7800000,7900000,8000000,8100000,8200000,8300000,8400000,8500000,8600000,8700000,8800000,8900000,9000000,9100000,9200000,9300000,9400000,9500000,9600000,9700000,9800000,9900000,10000000,10100000,10200000,10300000,10400000,10500000,10600000,10700000,10800000,10900000,11000000,11100000,11200000,11300000,11400000,11500000,11600000,11700000,11800000,11900000,12000000,12100000,12200000,12300000,12400000,12500000,12600000,12700000,12800000,12900000,13000000,13100000,13200000,13300000,13400000,13500000,13600000,13700000,13800000,13900000,14000000,14100000,14200000,14300000,14400000,14500000,14600000,14700000,14800000,14900000,15000000,15100000,15200000,15300000,15400000,15500000,15600000,15700000,15800000,15900000,16000000,16100000,16200000,16300000,16400000,16500000,16600000,16700000,16800000,16900000,17000000,17100000,17200000,17300000,17400000,17500000,17600000,17700000,17800000,17900000,18000000,18100000,18200000,18300000,18400000,18500000,18600000,18700000,18800000,18900000,19000000,19100000,19200000,19300000,19400000,19500000,19600000,19700000,19800000,19900000,20000000,20100000,20200000,20300000,20400000,20500000,20600000,20700000,20800000,20900000,21000000,21100000,21200000,21300000,21400000,21500000,21600000,21700000,21800000,21900000,22000000,22100000,22200000,22300000,22400000,22500000,22600000,22700000,22800000,22900000,23000000,23100000,23200000,23300000,23400000,23500000,23600000,23700000,23800000,23900000,24000000,24100000,24200000,24300000,24400000,24500000,24600000,24700000,24800000,24900000,25000000,25100000,25200000,25300000,25400000,25500000,25600000,25700000,25800000,25900000,26000000,26100000,26200000,26300000,26400000,26500000,26600000,26700000,26800000,26900000,27000000,27100000,27200000,27300000,27400000,27500000,27600000,27700000,27800000,27900000,28000000,28100000,28200000,28300000,28400000,28500000,28600000,28700000,28800000,28900000,29000000,29100000,29200000,29300000,29400000,29500000,29600000,29700000,29800000,29900000,30000000,30100000,30200000,30300000,30400000,30500000,30600000,30700000,30800000,30900000,31000000,31100000,31200000,31300000,31400000,31500000,31600000,31700000,31800000,31900000,32000000,32100000,32200000,32300000,32400000,32500000,32600000,32700000,32800000,32900000,33000000,33100000,33200000,33300000,33400000,33500000,33600000,33700000,33800000,33900000,34000000,34100000,34200000,34300000,34400000,34500000,34600000,34700000,34800000,34900000,35000000,35100000,35200000,35300000,35400000,35500000,35600000,35700000,35800000,35900000,36000000,36100000,36200000,36300000,36400000,36500000,36600000,36700000,36800000,36900000,37000000,37100000,37200000,37300000,37400000,37500000,37600000,37700000,37800000,37900000,38000000,38100000,38200000,38300000,38400000,38500000,38600000,38700000,38800000,38900000,39000000,39100000,39200000,39300000,39400000,39500000,39600000,39700000,39800000,39900000,40000000,40100000,40200000,40300000,40400000,40500000,40600000,40700000,40800000,40900000,41000000,41100000,41200000,41300000,41400000,41500000,41600000,41700000,41800000,41900000,42000000,42100000,42200000,42300000,42400000,42500000,42600000,42700000,42800000,42900000,43000000,43100000,43200000,43300000,43400000,43500000,43600000,43700000,43800000,43900000,44000000,44100000,44200000,44300000,44400000,44500000,44600000,44700000,44800000,44900000,45000000,45100000,45200000,45300000,45400000,45500000,45600000,45700000,45800000,45900000,46000000,46100000,46200000,46300000,46400000,46500000,46600000,4
```

```
NC_001751.1, Position,0,1
NC_001751.1,Count,0,0
```

```
NC_011033.1, Position,0,1
NC_011033.1,Count,0,0
```

[Here](#) you can find a tab-separated table.